

Zihan Yang

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Research Profile

Researcher at the intersection of control and machine learning, focusing on robust locomotion, disturbance estimation, and feedback-informed robot learning. Currently a research intern at Galbot working on humanoid locomotion and manipulation. Work includes two ICRA 2026 papers, an accepted *npj Robotics* article, an ICLR 2025 oral paper, and the ICCA 2024 Best Student Paper Award.

Education

Beihang University Sep 2023–Jan 2026

M.S. in Aerospace Science and Technology

- GPA: 3.80/4.00; supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo.

Northwestern Polytechnical University / Queen Mary University of London Sep 2019–Jun 2023

Joint NPU–QMUL B.Eng. dual-degree program in Materials Science and Engineering

- NPU GPA: 3.59/4.00; QMUL degree classification: First Class Honours.

Research Experience

Galbot Present

Research Intern

- Conduct research on humanoid locomotion and manipulation.

Beihang University Sep 2023–Jan 2026

Graduate Researcher; control and machine learning

- Unified Meta-Representation and Feedback Calibration:** Developed a framework to predict non-structural disturbances and adapt model parameters online while attenuating learning residuals.
- Learning-Based Observer for Coupled Disturbance:** Used Chebyshev polynomial approximation to decouple state-coupled and external-input-related disturbances and learn the disturbance model online.
- Feedback Favors the Generalization of Neural ODEs:** Co-developed feedback neural networks for continuous-time tasks, validated on trajectory prediction and quadrotor MPC.
- TRACE—Trajectory Refinement with Control Error:** Exploited quadrotor differential flatness for safe, accurate maneuvers without controller fine-tuning; won the ICCA 2024 Best Student Paper Award.

Beihang University / Unitree Robotics 2025

Quadruped robotics research

- Developed a teacher-student locomotion policy and worked on LiDAR-inertial odometry and mapping for quadruped robots.

Beihang University / China Academy of Launch Vehicle Technology 2024

Successive Convexification-Based Planning and Disturbance-Aware Tracking Control of Quadrotor Swarms

- Optimized collision-free multi-agent formation trajectories with successive convexification and designed disturbance-observer-based tracking control under wind disturbances.

Northwestern Polytechnical University 2021–2023

Selected undergraduate projects

- Finite Element Simulation of 3D-Printed Alumina Sintering Process:** Coupled porosity, heat-conduction, and sintering-stress models to assess stress overload caused by temperature gradients in large porous green bodies.
- Potential Flow Solver for Lift Coefficient Prediction Under Flow Separation:** Used CFD to capture separation and map lift loss to an equivalent decambered airfoil, then embedded the mapping in an iterative vortex-panel solver.
- Deep CNN for Airfoil Inverse Design:** Trained a ResNet-based model on 10,000 XFOIL-labeled airfoils to predict candidate geometries from target pressure distributions.

Publications

* Equal contribution.

Peer-reviewed publications

- [1] J. Jia*, S. Han*, M. Wang*, G. Li, **Z. Yang**, S. Zhou, K. Guo, J. Yang, X. Yu, W. Wang, and L. Guo, “Physics filtering favors the generalization of robot learning,” *npj Robotics*, 2026, Accepted.
- [2] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, “Unified meta-representation and feedback calibration for general disturbance estimation,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [3] J. Jia*, M. Wang*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, “Learning-based observer for coupled disturbance,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [4] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, “Flying in harsh environments: Anti-rain quantification system and control strategy,” *IEEE Transactions on Industrial Electronics*, vol. 72, no. 10, pp. 10 349–10 358, 2025.
- [5] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, “Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture,” in *2025 IEEE 19th International Conference on Control & Automation (ICCA)*, 2025, pp. 878–884.
- [6] J. Jia*, **Z. Yang***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, “Feedback favors the generalization of neural ODEs,” in *The Thirteenth International Conference on Learning Representations (ICLR)*, Oral paper (1.8% oral acceptance rate), 2025.
- [7] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, “TRACE: Trajectory refinement with control error enables safe and accurate maneuvers,” in *2024 IEEE 18th International Conference on Control & Automation (ICCA)*, Best Student Paper Award, 2024, pp. 154–161.

Manuscripts and preprints

- [8] **Z. Yang**, S. Han, K. Guo, and X. Yu, “CARO: Contact-agnostic residual observation for zero-shot robust quadruped locomotion,” 2026, Preprint.
- [9] K. Guo*, **Z. Yang***, Y. Liu, J. Jia, and X. Yu, “Optimizing control-friendly trajectories with self-supervised residual learning,” 2026, Under review.

Honors and Awards

Oral Paper (1.8% Oral Acceptance Rate) <i>The Thirteenth International Conference on Learning Representations (ICLR), Singapore</i>	Apr 2025
Best Student Paper Award <i>IEEE 18th International Conference on Control and Automation (ICCA), Reykjavík, Iceland</i>	Jun 2024
Second-Class Scholarship <i>Beihang University</i>	2024, 2023

Technical Skills

Programming: Python; C/C++.

Robotics/ML: Learning-based control; neural ODEs; meta-learning; LiDAR-inertial odometry/mapping.

Simulation/Control: MATLAB/Simulink; XFOIL; trajectory optimization; disturbance observers; LQR; PID.

Hardware: Fixed-wing, quadrotor, and quadruped systems; Pixhawk, Jetson Nano, and Intel NUC; 3D printing, laser cutting, and electronic assembly.

Languages: Mandarin Chinese (native); English (fluent, IELTS 7.5); Japanese (basic).